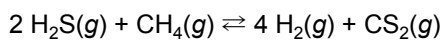
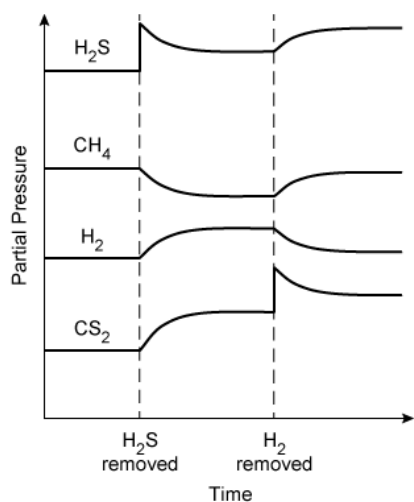


A sealed, rigid container is filled with a mixture of $\text{H}_2\text{S}(g)$, $\text{CH}_4(g)$, $\text{H}_2(g)$, and $\text{CS}_2(g)$. The mixture is initially at equilibrium as described by the reaction



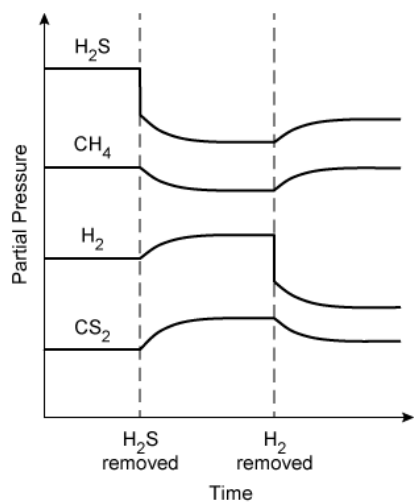
A researcher then perturbs the equilibrium by removing some $\text{H}_2\text{S}(g)$ from the mixture. The mixture is allowed to reestablish equilibrium, and then the researcher removes some $\text{H}_2(g)$. Which graph best represents the partial pressures of each gaseous species as a function of time? (Note: Assume the volume stays constant.)

☐ A.



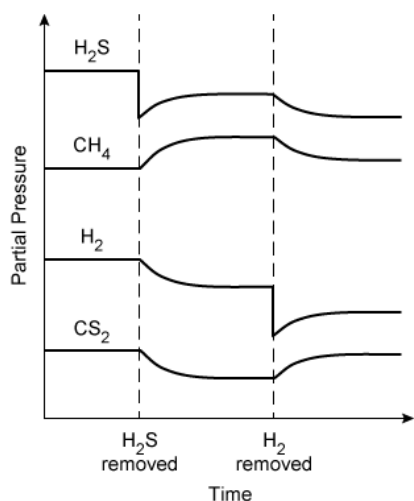
[4%]

☐ B.



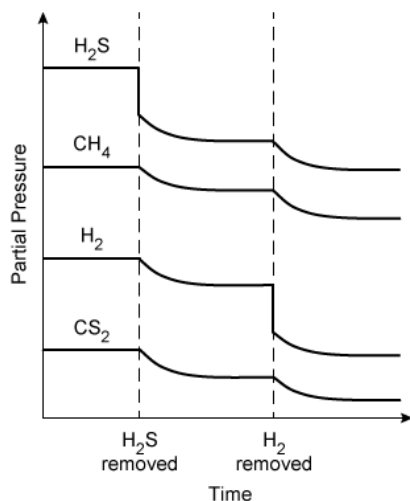
[16%]

☒ C.



[73%]

☐ D.

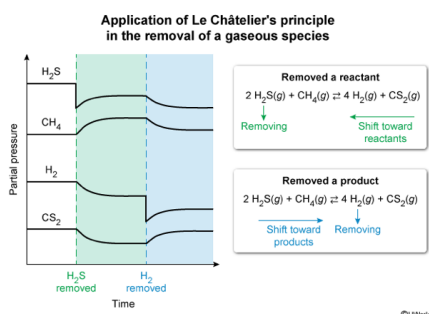


[6%]

Correct

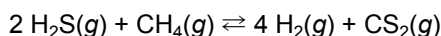
72% answered correctly

Explanation:



Le Châtelier's principle says that if the **equilibrium state** of a system is perturbed (eg, removal of a reactant or a product), the system will shift toward the reactants or toward the products to counteract the disturbance and establish a new equilibrium. **Removing products** causes the reaction to **form more products** (ie, the equilibrium shifts to the right), whereas **removing reactants** causes the reaction to **form more reactants** (ie, the equilibrium shifts to the left).

The question states that a mixture of the gases involved in the reaction



is initially at equilibrium. A researcher then perturbs the equilibrium by first removing $\text{H}_2\text{S}(\text{g})$ from the mixture; consequently, the partial pressure of $\text{H}_2\text{S}(\text{g})$ decreases sharply. This causes the reverse reaction to be favored (ie, the reaction shifts toward reactants). In response, $\text{H}_2(\text{g})$ and $\text{CS}_2(\text{g})$ react to form $\text{H}_2\text{S}(\text{g})$ and $\text{CH}_4(\text{g})$, causing gradual increases in $\text{H}_2\text{S}(\text{g})$ and $\text{CH}_4(\text{g})$ partial pressures along with gradual decreases in $\text{H}_2(\text{g})$ and $\text{CS}_2(\text{g})$ partial pressures.

After allowing the mixture to reestablish equilibrium (ie, the partial pressures become **constant** as a function of time), some $\text{H}_2(\text{g})$ is removed from the mixture. As such, a sharp decrease in the partial pressure of $\text{H}_2(\text{g})$ occurs. This causes the forward reaction to be favored (ie, the reaction shifts toward products). In response, $\text{H}_2\text{S}(\text{g})$ and $\text{CH}_4(\text{g})$ react and gradually decrease in partial pressure, causing the partial pressures of $\text{CS}_2(\text{g})$ and $\text{H}_2(\text{g})$ to gradually increase. Therefore, the graph shown in Choice C best represents the partial pressures of the gaseous species after each perturbation.

(Choice A) The changes shown in this graph would occur if $\text{H}_2\text{S}(\text{g})$ and $\text{H}_2(\text{g})$ were sequentially *added* to the mixture, not *removed*.

(Choices B and D) Removing $\text{H}_2\text{S}(\text{g})$ from the mixture should cause the reaction to shift toward *reactants* (ie, the partial pressures of $\text{H}_2\text{S}(\text{g})$ and $\text{CH}_4(\text{g})$ should gradually *increase*, and the partial pressures of $\text{H}_2(\text{g})$ and $\text{CS}_2(\text{g})$ should gradually *decrease*). Removing $\text{H}_2(\text{g})$ from the mixture should have the opposite effect.

Educational objective:

Le Châtelier's principle states that if a reaction mixture at equilibrium undergoes an alteration in reactant or product concentration, the system will shift in the direction that opposes the change.

Time spent:0 sec

Subject:
General Chemistry

QID:403892

System:
Thermodynamics, Kinetics & Gas Laws